UTKARSH RAJ

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PROFESSIONAL SUMMARY

Detail-oriented **Information Science & Engineering undergraduate** with strong foundations in Object-Oriented Design, Data Structures & Algorithms, and Database Management Systems. Demonstrated expertise in architecting full-stack web applications, event-driven real-time architectures, and modular RESTful APIs using **Java, Node.js, Express.js, MongoDB, and MySQL**. Experienced in designing low-latency communication pipelines, securing token-based authentication, and optimizing relational and NoSQL databases.

TECHNICAL SKILLS

Languages:	Java (Core Java, Multithreading, OOP), JavaScript (ES6+), SQL
Backend Development:	Node.js, Express.js, RESTful APIs, WebSockets (Socket.IO), JDBC
Frontend Development:	HTML5, CSS3, DOM Manipulation, Responsive Web Design
Databases & ORM:	MongoDB (Mongoose), MySQL, Database Normalization, Query Indexing
Core Computer Science:	Data Structures & Algorithms (DSA), DBMS, Operating Systems, OOP
Tools & Platforms:	Git, GitHub, VS Code, Render

TECHNICAL PROJECTS

Real-Time Collaborative Chat Application

Technologies: JavaScript, Node.js, Express.js, Socket.IO, MongoDB, HTML5, CSS3

GitHub: github.com/utkarshraj169/chat-application.git | **Live Demo:** chat-application-2-ftri.onrender.com

- Architected an event-driven messaging platform with **Socket.IO**, sustaining **50+ concurrent active connections** with **<80ms message broadcast latency**.
- Implemented secure token-based authentication using **JWT** and **bcrypt (10 salt rounds)**, reducing unauthorized access attempts to **0%**.
- Engineered optimized **MongoDB compound indexes** on timestamps and channel IDs, decreasing message query response times by **40%**.
- Developed **6+ RESTful API endpoints** in Express.js for room management, message history pagination, and user profile state.

Student Management & Analytics System

Technologies: Java, JDBC, MySQL, OOP Architecture

GitHub: github.com/utkarshraj169/student-management-system.git

- Developed an enterprise-style desktop records manager using **OOP patterns**, managing **1,500+ structured student records** with zero data corruption.
- Engineered a **JDBC Data Access Layer** with parameterized statements and connection pooling, boosting batch query throughput by **30%**.
- Automated data validation algorithms preventing duplicate enrolments and invalid formats across **10+ relational database fields**.
- Implemented normalized **MySQL tables (3NF)** with foreign key cascade rules, ensuring **100% ACID compliance** during multi-record transactions.

Personal Finance & Expense Tracker

Technologies: JavaScript, Node.js, Express.js, MongoDB, CSS3

GitHub: github.com/utkarshraj169/personal-finance-tracker.git | **Live Demo:** utkarshraj169.github.io/personal-finance-tracker/

- Engineered a personal finance tracking web app processing **500+ transaction entries** with instant mathematical net-balance calculations.
- Constructed multi-stage **MongoDB Aggregation Pipelines** to compute category summaries and monthly trends in **<50ms execution time**.
- Built **5 modular REST API endpoints** with centralized error-handling middleware, achieving **99.9% predictable JSON response uptime**.
- Integrated asynchronous client-side DOM rendering with vanilla JavaScript, eliminating **100% of full-page reloads** for CRUD interactions.

EDUCATION

National Institute of Engineering — Expected Graduation: June 2027 (Sept 2023 – 2027)

Bachelor of Engineering in Information Science and Engineering — **CGPA: 7.4 / 10.0**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), Database Management Systems (DBMS), Operating Systems, Computer Networks.